Project 2: Cryptography MATH 4175
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For m = 6:
Substring y1 =
LDXEJVWTLSNYZOLRQGTVSWQGPJXPWVRHKCKLHWPRASULXYFMKGWGTEBDQGRA
FBKSLVDPZCPPQLJMJUQRZVEAUBVPEWAZVICGJASQRWEENLORTYCEGLILEQOMLIOE
DAWLIOTOPUEWLIGKSYPOJWDWLRDEEDKWHZYMEGVAYZZAJJEPIIYSVSFVBJLYXFDJP
SHAYXWJJEFZSXPLHCSNCJ
Index of Coincidence for y1: 0.040920483025746184

Substring y2 =
DJRMMUTALSXPKMUKZIPKMDSDREIANVHERJTNJVMLLMAKLECIVBJRGDYVLWLSLFZQ
RUWLRMWSPXPJKAULRQZSALBGTETAGQXAFVPERITFIKWLGPEWRKGAEEMJGSIZZLYL
VXSPPWWVRONBJWHXNEQTNFXRYMJTHRPVAJLLKYPUEJQVLGPZWIPNZKDWKWFREH
WHIZELIDDYPITUULSDIW
Index of Coincidence for y2: 0.04251537935748462

Substring y3 =
MPXFIHNKPJLPJTWAWLRSMIJDPVZRDAVBMPVWWLUWWEUOEEIEKMKEDJPFZNFQGC
UBASXJWWUJMLVYIZKUVEEVAKHRGDPPBVKPEKIKNFGCKVYUHGTEQOODWRTQJJUX
ORIDAKRDRVWWBTPYGNLKYDEMAZLSTIWAUZVVCESLUHLZDKVVJAKLOLWSYVIXWRG
EVTPOHMPZMELWWZZIQFXEEC
Index of Coincidence for y3: 0.04155844155844156

Substring y4 =
UUFWHMLFMKFLEFWEOFWWPMMASHHYZAELYVNCKKVDXQVWADFLPCVILUXWFMNK
RHFYLJWHNZGGJGYSJKUWTXLJJLPEDWTRBQALLSZBALQOCRBZARGDUMEJHXZTAKU
SIELICWRYEEKZLDHLJJOWVAETJWDLPPKMEZWPLVWDWXGNRCKLWVSIIDAPKCKMBX
KINRPEVQWYUKAZWXMEZXJQIZ
Index of Coincidence for y4: 0.04160401002506266

Substring y5 =
PVCDFOCXUBCTQVEYMNEKVPNAESXHNZVDWVSZTBWJOSYDAAGHRPZCEGHSGMMA
MHPZNBZVXVWTQJAZTLDBUDWRLVBILRRYYAOWZIDZZNWVIYIMBEDTPIBNWKQLKDWJ
WEDJVPDFMZTJDJRZTLIZKLLFIZHFHLJEAAGHXXIKOKDWNGBKSREPCMLHFRSVWOVE
ERRTKYWJAUIKLHIOZWHVFK
Index of Coincidence for y5: 0.03886990202779676

Substring y6 =
TYJIWDVQEVBAVLLZIMKGZSDVVIUVIRRIOKQZPYBASUXPLBRUWGNEWIUJMPJBNJBVLZ
JAVHIZTAYWWMFZLPZHVNYCEDWHPVIOZGMZVWPGVIZMBILUVGYIGGPXSZVKZEIOVM
KZWXPPQPFHPWEQELHXLVWCLDILZBOYLYVFAPVSCDYKJSWWWDTAGVEYTWAMQVE
CVDSRYPSEKUUWEKIKZB

Index of Coincidence for y6: 0.04529505582137161

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For m = 7:
Substring y1 =
LJXWFDTLJFTVLKWFEGWSDSSUWVVLWKLJLVJSUKEDGUGJELGUQWFKMJSRSWVVPS
MGAWUUUTDZULHELDZGVAWZSENLWGLWUAELGKOJWGMGJUJZWLAWDZUWWLJFSW
LWKLWNZDFLKTUZGYGLLWKVJJVLSSSWLAHGLWWXVMSWOVYSJDLZLULUFJVZ
Index of Coincidence for y1: 0.071260086902545

Substring y2 =
DPFDWWAPKCAOUAONKSDJAEIPNAEDOKNWKWASAOAARKBKIEIDLNNANKQAJZACWJ
JJYJAKWUPAAKPIEAABQOOAPKANPNKYZBIEROENGOJJKWKAYDCPKPWWZDPKJNOZE
DTAWHCDJAEAOEJSDOPAEKKKJYZODLAJDXBOAPHPEKDJIEAKKPUQXHKJ
Index of Coincidence for y2: 0.071260086902545

Substring y3 =
MUCIVTKMBBZMWEMMVMIMAVXADAVICTWKBBAMUWABMVMVCWBVZMMBBZBLBJZMU
GQAMKZUBLESALBCWTPBAIJVIBZWEIVBMBCWQMBIQMQADVDLIIVMOPVKJQGBGJIQW
QMJZWEMWMABMAEWKAZUDCBKIPLIMTBKIMWWJETPTVWLMKIEXTIZWICW
Index of Coincidence for y3: 0.06815642458100558

Substring y4 =
UVJJUNFUVYKTWYITKMMNVJIRZZRKJVCTYRLEVDLFIKCZEEYFFMJFFUYNZPRWGTTJJI
KDZVZVJVYEEPRYVGFKZZVEFKRIMYEEUIYEETTKZEZRLJVTPRETPINYJLEJEETIVEYIKE
ZYVCVIFZPZRGYIGKICDVZVKVTDRVRRCXEZUUSSIZEZKNIC
Index of Coincidence for y4: 0.061390440720049656

Substring y5 =
PYEMHLXENPJFEZGPSPPDPEZYNRHRPNZPPLWQYPYCEPPNTDPWGPALCFZLDLWZWZ
LPYJLFZQEJLNPTDTYPCAESDZWTCCYZTPTDPGLERZLSOZOEDOOSDEZPLOPLTWOND
EFLDRTPJLZPVLXVYYLNNDPLASPWFNYCSYFFENREYZPYAPZPZMOESDEZ
Index of Coincidence for y5: 0.06834264432029795

Substring y6 =
TXMIMCQSXPEVLQIRWVSGRVHHIREMVSZWMWXSXXEILRGGGJXSMRSGHPVVWJNVIQ
XVSTMRRELRVVGGWRHIXPLIMRIGOIIRGGGTVIAWXQXIIXIEIIXRYMXWRTHZPPXYALXRX
SPLIHRVPXYAKHGWCEVJVEWSPSKREXWGIEVAIMWJYFYWXIWCSEIK
Index of Coincidence for y6: 0.06225946617008069

Substring y7 =
DRFHOVLSLLQLRZLWKZQDPHXVVHBYVQHVUDOULLEFHWWRDUHJGLQRHBLUXHXHP
PLYZWQLVXWHBBRDRWVQKLZGQRFQVVOLHRDULGDHKPLSUSWELVKRFWEVBDRHYH
KVLHLFLLHDWHZWHLVLUXDSJQVWRWVIWPFVYKRKEQHHHQWREDWWHUHLXQFB
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Index of Coincidence for y7: 0.06790813159528243

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For m = 8:
Substring y1 =
LPCJHCLJCZTEQLESINPVXWAVKPSHLWAEYXEGKMZTJHQNMFCPLSZZWWQLAJZDZEW
UKBEDRVVOJKDRFWNVITGDGOBERQLJWDRDIKDPVTLTRSNIJDLLZHEIJHZGEEIYHDJK
BIAEVLLBVSXRVPTRYMWJEISZIHFHCC
Index of Coincidence for y1: 0.04160283802305894

Substring y2 =
DUJMMVLKBKFLZFKMMDRHUNARRVQJKBLQXLDRVCNGUULMJLHBRJJRZIPGYKKFRXZ
ALYTWWGQIFSMRLPIRZGRLRMYEXPGKVZEIVWKPEPRDFJJENAHNWWYPIHZOAVVKXV
ECYLVWWDTZCEKXAEREIQSIKSPXUUXIIZ
Index of Coincidence for y2: 0.040635330162057566

Substring y3 =
MVEIOTPBYJVRWNVMPGPSPDZHMVLWBRWSLEAMKPGDGDZMAGHSABPWVPMJMILRV
DAAVPGRZBAGEIQNNEKYRHEEQILWKMJDEOELAPORZWBJKGLOYLWAZETLWUAMCXA
UKADGPJRSOMVYRYWOJVRAHYJMUZWHLQWNEK
Index of Coincidence for y3: 0.04377973071031202

Substring y4 =
UYMHDAMVPELKOMKPSDSIAZREYKNKYLXUKABIPGRLIVFPSRJQLZLNHSJAJJMLTPSJN
GDDABVALGEAWFCILAIWUGAHGJAZZIELCMPEXVLPBJWXVLTJVRPDTEBVLYLWPURDV
WSZIDNKVWBWRNVHVDLUPYWUUZKDIB
Index of Coincidence for y4: 0.04248972022897686

Substring y5 =
PXFFWKUNPQLAMTSVQDEXRNRBWKWTPWOUOAFERWEEBFGRQMKBNDJXPJQJYTQU
UEVLVRLAPYCPZSKZECIOUBCEPIDWOQKOXWWDVTDMEWDGYTPKKDMIDSHKAAYVXV
LOZZNEVSYLCFSFLXWDEEHOKWZAFWLPIZSEF
Index of Coincidence for y5: 0.03927813163481953

Substring y6 =
TRWWTFEXLVUEIPWZSAVIYIHLOTCPMDSAWLCLWJIWYWMLKNZYLWHVWGTPSWUWLZ
JVBEETRPXLZPBVTOVWZBEDVGJGMTSISZYIVSYWWZQNLPHWEQTLXLLJMZPPLLDAPN
CQLJPIWPPGDMTFIQWEVEYYDZKTEESQZ
Index of Coincidence for y6: 0.05038379879144211

Substring y7 =
DXDVNXSLTOWYGRKWJAJZHVVDCVZWUJSUDYIHGKCEPSGFABUZVXVCUTLVZUKBVER
BHIWPYIKWAIZWGVLYMYTTLONQTLIUJAIJORFUWJIPZYLZWEFRLFDWEZVHGSKZLWJV
KIKPSWHJIVFGESPTXPJELKXZOCXVJ
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Index of Coincidence for y7: 0.041401273885350316

Substring y8 =
JFIULQSFAMWZIWGDMVEHVVEIJNZVVAMVPEFUBVEDXJWNBFFVUWAMGZXYWAUZQLH
LPCETHQAOVZZIQGKBMPGUKEIEZXSUKLLOXRZWKPOHHWOQEEXFDCMKLRWYJWFY
GSJKKGSWIAAKKYWKMHPCZWRDAEIMWLJKW
Index of Coincidence for y8: 0.04131961456802221

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Our outputs verify that m = 7 is the correct guess for the length of the keyword. For m = 6 and m = 8, the indices of coincidence for each substring are all somewhat close to 0.038, which is the expected index of coincidence for a completely random substring. By contrast, for m = 7, all the substrings have indices of coincidence that are close to or exceeding 0.065, which would be the index of coincidence for a substring with letter frequencies approaching those of the English language as a whole. Thus, this indicates that each of the substrings for m = 7 is likely to be a ciphertext encrypted with the same key, which would be expected for a ciphertext encrypted with the Vigen�re cipher with a keyword of length 7.

Dot Product Table (rows = shifts 0-25)

0:	3.314	4.508	3.422	4.090	4.248	4.019	3.373
1:	3.614	3.314	3.880	3.859	3.815	3.186	3.036
2:	3.969	3.767	3.413	4.276	3.379	3.835	3.500
3:	4.934	3.538	3.278	3.325	3.386	4.152	6.558
4:	4.052	3.133	4.527	4.402	3.626	6.372	4.270
5:	4.151	3.375	3.536	3.956	3.533	3.829	3.141
6:	3.834	4.318	3.142	4.606	3.354	3.496	2.887
7:	4.912	4.436	3.701	3.658	4.764	3.143	5.094
8:	3.793	3.779	6.663	3.630	3.513	4.254	3.502
9:	3.389	4.233	3.926	2.939	3.239	3.454	3.554
10:	3.046	4.233	3.223	3.506	3.787	3.655	3.496
11:	4.056	4.078	2.962	4.212	6.717	3.468	3.859
12:	3.061	3.804	4.102	3.654	3.906	3.116	3.227
13:	3.362	3.332	3.183	4.246	3.522	3.958	3.917
14:	4.066	3.084	3.714	3.031	3.009	4.001	4.319
15:	3.546	3.728	3.809	3.504	4.069	4.771	3.619
16:	3.688	3.612	3.678	4.118	2.860	3.624	4.022
17:	4.248	3.649	3.758	6.282	3.801	4.424	4.112
18:	6.618	4.406	4.053	3.562	3.937	3.890	4.708
19:	3.749	3.437	4.676	3.193	3.180	4.598	4.066
20:	3.339	2.836	3.887	3.519	3.486	3.549	3.636
21:	3.266	4.022	4.238	4.646	4.101	3.667	2.984
22:	3.996	6.834	3.836	3.423	4.826	3.192	3.521
23:	2.976	4.167	4.396	3.369	3.907	3.522	4.042
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24: 3.503 3.106 3.760 3.701 4.293 3.581 3.535
25: 3.619 3.371 3.337 3.393 3.842 3.346 4.123

Decrypted Text:
THEDEPARTMENTOFJUSTICEHASBEENANDWILLALWAYSBECOMMITTEDTOPROTECTIN
GTHELIBERTYANDSECURITYOFTHOSEWHOMWESERVEINRECENTMONTHSHOWEVER
WEHAVEONANEWSCALESEENMAINSTREAMPRODUCTSANDSERVICESDESIGNEDINAW
AYTHATGIVESUSERSSOLECONTROLOVERACCESSTOTHEIRDATAASARESULTLAWENFO
RCEMENTISSOMETIMESUNABLETORECOVERTHECONTENTOFELECTRONICCOMMUNIC
ATIONSFROMTHETECHNOLOGYPROVIDEREVENINRESPONSETOACOURTORDERORDU
LYAUTHORIZEDWARRANTISSUEDBYAFEDERALJUDGEFOREXAMPLEMANYCOMMUNICA
TIONSSERVICESNOWENCRYPTCERTAINCOMMUNICATIONSBYDEFAULTWITHTHEKEYNE
CESSARYTODECRYPTTHECOMMUNICATIONSSOLELYINTHEHANDSOFTHEENDUSERTHI
SAPPLIESBOTHWHENTHEDATAISINMOTIONOVERELECTRONICNETWORKSORATRESTO
NANELECTRONICDEVICEIFTHECOMMUNICATIONSPROVIDERISSERVEDWITHAWARRAN
TSEEKINGTHOSECOMMUNICATIONSTHEPROVIDERCANNOTPROVIDETHEDATABECAUS
EITHASDESIGNEDTHETECHNOLOGYSUCHTHATITCANNOTBEACCESSEDBYANYTHIRDP
ARTYWEDONOTHAVEANYSILVERBULLETSANDTHEDISCUSSIONSWITHINTHEEXECUTIV
EBRANCHARESTILLONGOINGWHILETHEREHASNOTYETBEENADECISIONWHETHERTOS
EEKLEGISLATIONWEMUSTWORKWITHCONGRESSINDUSTRYACADEMICSPRIVACYGRO
UPSANDOTHERSTOCRAFTANAPPROACHTHATADDRESSESALLOFTHEMULTIPLECOMPE
TINGCONCERNSTHATHAVEBEENTHEFOCUSOFSOMUCHDEBATEBUTWECANALLAGREE
THATWEWILLNEEDONGOINGHONESTANDINFORMEDPUBLICDEBATEABOUTHOWBESTT
OPROTECTLIBERTYANDSECURITYINBOTHOURLAWSANDOURTECHNOLOGY

Decrypted Text Formatted with Spaces and Punctuation:
The department of justice has been and will always be committed to protecting the liberty and security of those whom we serve. In recent months, however, we have on a new scale seen mainstream products and services designed in a way that gives users sole control over access to their data. As a result, law enforcement is sometimes unable to recover the content of electronic communications from the technology provider even in response to a court order or duly authorized warrant issued by a federal judge. For example, many communications services now encrypt certain communications by default with the key necessary to decrypt the communications solely in the hands of the end user. This applies both when the data is in motion over electronic networks or at rest on an electronic device. If the communications provider is served with a warrant seeking those communications the provider can not provide the data because it has designed the technology such that it can not be accessed by any third party. We do not have any silver bullets, and the discussions within the executive branch are still ongoing. While there has not yet been a decision whether to seek legislation, we must work with Congress, industry, academics, privacy groups, and others to craft an approach that addresses all of the multiple competing concerns that have been the focus of so much debate. But we can all agree that we will need ongoing, honest, and informed public debate about how best to protect liberty and security in both our laws and our technology.